

Reviewed Preprint

v1 • January 19, 2026

Not revised

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Competing interests: No competing interests declared

Funding: See [page 21](#)

Reviewing editor: Dominique Ferrandon, Institut de Biologie Moléculaire et Cellulaire, France

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Reproducibility of Scientific Claims in *Drosophila* Immunity: A Retrospective Analysis of 400 Publications

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eLife Assessment

This **fundamental** study is part of an impressive, large-scale effort to assess the reproducibility of published findings in the field of *Drosophila* immunity. In a companion article, the authors analyze 400 papers published between 1959 and 2011, and assess how many of the claims in these papers have been tested in subsequent publications. In this article, the authors report the results of validation experiments to assess a subset of the claims that, according to the literature, have not been corroborated. While the evidence reported for some of these validation studies is **convincing**, it remains **incomplete** for others.

<https://doi.org/10.7554/eLife.108404.1.sa4>

Abstract

Drosophila immunity has been the focus of intense study and has impacted other research fields including innate immunity and agriculturally or epidemiologically relevant investigations of insect pests and vectors. Unsurprisingly for such a large body of work, some published results were later found to be irreproducible. Although some results have been contradicted in the literature, many have no published follow-up, either due to a lack of research or low motivation to publish negative or contradictory results. We have addressed this by performing a reproducibility project that analyses the verifiability of claims from articles published on *Drosophila* immunity before 2011. To assess reproducibility, we extracted claims from 400 articles on the *Drosophila*

of this analysis was to centralize evidence on insights and findings to improve clarity for scientists that may base research programs on these data. All our data are published on a publicly available website associated with this article (<https://ReproSci.epfl.ch/>) that encourages community participation. This article provides a short summary of claims that were found to have contradictory evidence, which may help the community to assess past findings on *Drosophila* immunity and improve clarity going forward.

Introduction

Innate immune mechanisms have become central to our understanding of immunology (Danilova, 2006; Medzhitov and Iwasaki, 2024; Pradeu et al., 2024), although their study was long overshadowed by interest in adaptive immunity. Innate immune mechanisms are now widely recognized as having key importance as a first layer of defense, in the recognition of infectious agents, and in subsequent polarization of the immune response. These responses are activated by a specific set of receptors that sense microbe-specific molecules, virulence factors and/or infection-induced damage. Complex signaling pathways integrate this information to induce specific innate immune modules adapted to the characteristics of the encountered pathogen. Characterization of the innate immune response has greatly benefitted from the use of model organisms such as *Drosophila*, which is amenable to genetic studies and relies exclusively on innate immunity for defense (Buchmann, 2014; Westlake et al., 2024).

The *Drosophila* immune system is composed of several immune modules, some of which are conserved in vertebrates (Buchon et al., 2014; Liegeois and Ferrandon, 2022; Westlake et al., 2024). The systemic immune response is best characterized by the production and secretion into the hemolymph of host-defense peptides by the fat body and hemocytes. The best-characterized of these effectors are the antimicrobial peptides (AMPs). AMPs are small cationic peptides with antimicrobial activity, whose expression is induced to incredibly high levels in response to infection (Hanson and Lemaitre, 2020). The systemic antimicrobial response is predominantly regulated at the transcriptional level by the Toll and Imd signaling pathways. In *Drosophila*, the Toll pathway is activated by microbial cell wall components (fungal glucans and Lysine-type peptidoglycan), and microbial proteases. Toll signaling culminates in the activation of two NF- κ B transcription factors, Dif (dorsal-related immunity factor) and Dorsal, which regulate a large set of immune genes such as the Bomanins, the antifungal peptide Drosomycin, and many other proteins (Clemmons et al., 2015; De Gregorio et al., 2002; Troha et al., 2018). Flies lacking Toll pathway activity are viable but highly susceptible to infection by Gram-positive bacteria and fungi (Lemaitre et al., 1996). The Imd pathway is activated by DAP-type peptidoglycans produced by Gram-negative bacteria and a subset of Gram-positive bacteria (e.g. *Bacillus*) (Kaneko et al., 2004; Leulier et al., 2003). Binding of peptidoglycan to receptors of the Peptidoglycan Recognition Protein family (PGRP-LC, PGRP-LE, PGRP-SD) initiates an intracellular signaling cascade with homology to the TNF α -Receptor pathway, which ultimately activates the NF- κ B factor Relish (Choe et al., 2002; Gottar et al., 2002; Iatsenko et al., 2016; Takehana et al., 2004). The Imd pathway regulates the expression of many genes encoding antibacterial peptides, serine proteases, and the iron sequestration protein Transferrin, among others. Imd deficient flies are viable, but susceptible to Gram-negative bacterial infection.

Complementary to these transcriptional programs, melanization is an arthropod-specific immune response resulting in rapid deposition of the black pigment melanin at wound or infection sites and concomitant production of microbicidal reactive oxygen species (Cerenius et al., 2008). Melanization relies on the activation of enzymes called prophenoloxidases (PPOs), which catalyze the oxidation of phenols, resulting in melanin polymerization (Dudzic et al., 2015). This elegant process also hardens clots with melanin polymer plugs that prevent blood loss, akin to the mechanical function of fibrin scabs. Clotting, which involves both humoral factors and hemocytes, contributes to wound healing and has been implicated in host defense against nematodes (Theopold et al., 2014; Yadav and Eleftherianos, 2018).

In addition to soluble immune effectors, cellular components also play a crucial role in immunity. In adult flies, two hemocyte types known as plasmatocytes and crystal cells are present in circulation or as sessile cells attached to the inner cuticle (Banerjee et al., 2019 [DOI](#); Lanot et al., 2000 [DOI](#)). Plasmatocytes are macrophage-like cells with phagocytic activity, while crystal cells produce and store PPO enzymes and contribute to melanization. Use of hemocyte deficient flies has shown that hemocytes contribute to survival upon systemic infection to certain bacterial species, and several phagocytic receptors have been identified (Charroux and Royet, 2009 [DOI](#); Defaye et al., 2009 [DOI](#); Melcarne et al., 2019a [DOI](#); Stephenson et al., 2022 [DOI](#); Ulvila et al., 2011 [DOI](#)). Natural infection with microbes also triggers a local immune response in epithelia, which has been extensively characterized in the gut (Buchon et al., 2013 [DOI](#); Tafesh-Edwards and Eleftherianos, 2023 [DOI](#); Westlake et al., 2024 [DOI](#)). In this organ, a range of defenses including the peritrophic matrix, an acid pH, peristalsis, and epithelial repair through stem cell proliferation all contribute to deterring ingested pathogens. Local production of AMPs and ROS provide two complementary inducible defense mechanisms in the gut (Liehl et al., 2006 [DOI](#); Ryu et al., 2006 [DOI](#)). It is now clear that each organ contributes individual input to the immune response in an integrated manner, which vary according to the route of infection and the nature of the pathogens (Westlake et al., 2024 [DOI](#)). Finally, extensive studies have highlighted the role of the immune system upon mating and in the reproductive organs to prevent infection (Schwenke et al., 2016 [DOI](#); Siva-Jothy, 2009 [DOI](#)).

Drosophila immunity has been a topic of intense research, with thousands articles and reviews published on this topic (Westlake et al., 2024 [DOI](#)). Characterization of the *Drosophila* immune system has contributed significantly to fields including innate immunity and agriculturally or epidemiologically relevant investigations of insect pests and vectors. As expected in such a large body of work, some published results were later found to be irreproducible. Although some of these results have been contradicted in the literature, many have no published follow-up, either due to a lack of research or low motivation to publish negative or contradictory results. We have addressed this by performing an extensive reproducibility project that analyses the validity of claims from articles on *Drosophila* immunity accepted for publication before 2011. This time frame allows opportunity for publication of follow-up results. We extracted major and minor claims from 400 articles on the *Drosophila* immune response to bacteria and fungi and performed verification where possible by comparing to other published literature in the field. We also experimentally tested 55 ‘unchallenged’ claims with no published follow-up, a process that will be continuously expanded with the help of the community.

The intent of this analysis is not to assign merit to individual scientists or publications, but rather to centralize evidence on results or groups of results to improve clarity for scientists that may base their research on these data. Although conflicting data are often discussed at conferences or at less formal meetings, diffusion of this knowledge is limited due to lack of publications. Our hope is to provide a dataset with community-sourced proofs that allow a wider range of scientists to easily identify claims with contradictory evidence. In association with this article, we have published the *ReproSci* website, which describes and presents assessments of claims from 400 articles and encourages community participation (<https://ReproSci.epfl.ch/> [DOI](#)). This article synthesizes the main results on claims of particular interest discovered during the reproducibility analysis, while the website provides a more complete analysis. These include claims published in high profile journals that have been contradicted or have tenuous support. This article is accompanied by 28 supplementary materials that include new experimental data from various laboratories that support or contradict some unresolved claims that are discussed in this article. In many cases, re-analysis of these claims is made possible by advances in understanding of immune mechanisms and technology since publication of the initial articles. Statistical analysis of this reproducibility study and metascience insights obtained from this approach are discussed in a companion article.

Results

Overview of the reproducibility project

400 research articles focusing on *Drosophila* immunity published between 1959 and 2011 were selected by PubMed keyword retrieval and manual curation (Supplementary Table 1). Articles dealing with population genetics, virulence of entomopathogens, and the immune response to viruses and wasps were excluded. The main (1 per article), major (2-4 per article) and minor (0-4 per article) claims were extracted along with a summary of evidence supporting the claims and the methods of each article. The reproducibility of each claim was then assessed by analyzing subsequent publications on the topic from the same or other laboratories. Claims were assessed as verified, unchallenged, challenged, mixed, or partially verified (see Methods). All article summaries and assessments were released to the community on July 11th, 2023, via the *ReproSci* website, which allows users to comment on the claims and provide confirmatory or contradictory evidence. 45 major and 11 minor claims (many of which are discussed in the Supplement) were also experimentally tested in various laboratories using alternate approaches. For instance, a result initially obtained using RNAi may be tested using newly generated CRISPR mutants. Our project focuses on assessing the strength of the claims themselves (inferential/indirect reproducibility) rather than testing whether the original methods produce repeatable results (results/direct reproducibility). Thus, our conclusions do not directly challenge the initial results leading to a claim, but rather the general applicability of the claim itself. This article presents some of the main observations from the *ReproSci* project and is intended to allow the reader to quickly grasp key claims in the *Drosophila* immunity field with low reproducibility (challenged claims) or not as simple as proposed in the initial articles.

Disclaimer

The aim of the *ReproSci* project is to assess the replicability of claims made in articles published in the field of *Drosophila* immunity. This is in no way an assessment of the ‘scientific value’ of the research. Although a lack of replicability is often viewed negatively, papers that include non-replicable observations may nevertheless have a positive impact by developing new techniques, introducing interesting concepts, providing a foundation for future discoveries, or presenting results that are useful despite incorrect interpretation. Similarly, articles that include only well-supported claims may be of low scientific value if they only confirm previous observations and present little in the way of innovation. The leading authors (H.W., B.L.) acknowledge that the selection of claims within papers and assessments of those claims cannot be fully objective. Opening the *ReproSci* database to comment by community members is intended to provide a greater degree of objectivity and consensus and generate ongoing discussion beyond initial assessments.

Insights from the *ReproSci* project on the Toll pathway

Toll is not regulated by 18-wheeler

Following the characterization of the Toll receptor (Toll-1) as a key regulator of *Drosophila* AMP genes (Lemaitre et al., 1996 [DOI](#)), the Toll homolog 18-wheeler (Toll-2) was reported to have a complementary role in LPS sensing and regulation of the antibacterial response, similar to mammalian TLR4 (Williams et al., 1997 [DOI](#)). This paper was initially widely cited, but was directly challenged by a follow-up study revealing that 18-wheeler does not have a direct role in immunity, but rather indirectly affects survival and AMP gene expression through a general negative impact on fly development (Ligoxygakis et al., 2002 [DOI](#)).

GNBP1 is not a sensor of fungal glucan

GNBP1 (Gram-Negative Binding Protein 1) was initially characterized for its ability to bind LPS and fungal glucan, based on overexpression in cell culture and *in vitro* binding studies (Kim et al., 2000 [DOI](#)). Subsequent genetic studies revealed that this secreted protein instead cooperates with PGRP-SA in sensing of Gram-positive bacteria, and find no evidence that GNBP1 acts as an immune

sensor for LPS or glucan (Gobert et al., 2003 [↗](#); Pili-Floury et al., 2004 [↗](#)). GGBP3, a *Drosophila* homolog of GGBP1, does have glucan binding activity and activates immune responses to fungi *in vivo* (Gottar et al., 2006 [↗](#)).

GNBP1 does not process peptidoglycan

Gram-positive bacteria stimulate the Toll pathway through cooperative action of PGRP-SA and GGBP1, which activate the apical serine protease ModSP upon sensing of peptidoglycans (Buchon et al., 2009b [↗](#); Gobert et al., 2003 [↗](#); Pili-Floury et al., 2004 [↗](#)). Structural studies revealed that PGRP-SA can bind peptidoglycan, consistent with its demonstrated role as pattern-recognition receptor *in vivo* (Chang et al., 2004 [↗](#)). However, one study suggested that GGBP1 can also digest peptidoglycan, and that this hydrolytic function of GGBP1 is required for full activation of PGRP-SA and ModSP (Wang et al., 2006 [↗](#)). Reported enzymatic activity of GGBP1 has never received follow up. Studies in other insects such as *Tenebrio molitor* and *Manduca sexta* suggest that the primary function of GGBP1 is as an adaptor between PGRP-SA and the apical serine protease ModSP (Kim et al., 2008 [↗](#); Park et al., 2007 [↗](#); Wang et al., 2022 [↗](#)). *In vitro* experiments performed as part of *ReproSci* found that GGBP1 does not process peptidoglycan in the absence or presence of PGRP-SA (**Supplementary S1**). Similarly, pre-incubation of GGBP1 with peptidoglycan did not enhance ModSP autoactivation. These results indicate that contrary to the initial claim, GGBP1 has no enzymatic activity against peptidoglycan. The initial results may have been obtained due to residual impurities in preparations of recombinant GGBP1.

Spirit, Sphinx and Spherioide are not key components of the Toll pathway

An RNAi screen identified five serine proteases involved in the activation of the Toll ligand Spatzle during the immune response (Kambris et al., 2006a [↗](#)). The roles of two of these genes, *SPE* and *grass*, were confirmed in subsequent studies, although *grass* was shown to affect the whole pattern recognition branch of the Toll pathway and not only the PGRP-SA/GGBP1 branch as suggested in the initial study (El Chamy et al., 2008; Jang et al., 2006 [↗](#)). New data from *ReproSci* using viable loss-of-function mutations and recent-generation RNAi reveals that of the remaining three genes, neither the serine protease homologs Spherioide and Sphinx1/2 nor the serine protease Spirit are strictly required for Toll activation (**Supplementaries S2, S3, S4**). It is likely that Sphinx1/2 has no role in the immune response, while the immune-inducible gene Spirit may have a redundant or a subtle role. An independent study suggests Spherioide has a subtle role in Toll activation (Patnogi and Leclerc, 2017 [↗](#)); experiments performed for *ReproSci* support these results. Non-replicable results on the roles of Spirit, Sphinx and Spherioide in Toll pathway activation may be due to off-target effects common to first-generation RNAi tools (Ueda, 2001 [↗](#)).

The Toll immune pathway is not negatively regulated by wntD

Gordon et al., (2005a) [↗](#) found that *wntD* acts as a feedback inhibitor of *dorsal* but not *dif* during the innate immune response. *wntD* mutants had elevated expression of *Diptericin* but not *Drosomycin* in response to septic infection with the Gram-positive bacterium *Micrococcus luteus*, and increased susceptibility to *Listeria monocytogenes* that was rescued in *wntD*, *dorsal* double mutants. However, data from FlyAtlas (Thurmond et al., 2019 [↗](#)) show that *wntD* is only expressed at the embryonic stage and not at the adult stage at which the experiments were performed by (Gordon et al., 2005a [↗](#)). Using an independent null mutant of *wntD*, we did not observe any impact of *wntD* on expression of *Diptericin* and *Drosomycin*, or increased sensitivity to *Listeria monocytogenes* infection (**Supplementary S5**). Thus, we conclude that *wntD* is not a negative regulator of the Toll pathway during the immune response. The initial claim concerning *wntD* may be explained by a genetic background effect independent of *wntD*.

Toll is not activated by overexpression of GGBP3 or Grass

Experiments performed for *ReproSci* find that contrary to previous reports, overexpression of *GGBP3* (Gottar et al., 2006 [↗](#)) or *Grass* (El Chamy et al., 2008) in the absence of immune challenge does not effectively activate Toll signaling (**Supplementaries S6, S7**).

Insights from the reproducibility project on the Imd pathway

PGRP-SD is a component of Imd and not Toll signalling

PGRP-SD was initially identified as a secreted pattern-recognition receptor involved in the Toll pathway in response to certain Gram-positive bacteria (Bischoff et al., 2004 [DOI](#); Wang et al., 2008 [DOI](#)). This was inconsistent with subsequent observations that **i**) PGRP-SD binds DAP-type peptidoglycan found in Gram-negative bacteria which typically activate Imd signaling (Basbous et al., 2011 [DOI](#); Leone et al., 2008 [DOI](#); Lim et al., 2006 [DOI](#)); and that **ii**) this gene is regulated by the Imd pathway (De Gregorio et al., 2002 [DOI](#)). Using isogenic fly stocks and newly generated mutants, Iatsenko et al., subsequently showed that PGRP-SD is a positive regulator of the Imd pathway upstream of PGRP-LC that promotes sensing of DAP-peptidoglycan (Iatsenko et al., 2016 [DOI](#)). Analysis of several fly stocks expected to carry the *PGRP-SD*^{ΔS3} mutation used in the initial study revealed the presence of a wild-type copy *PGRP-SD*, suggesting that either the stock used in this study did not carry the expected mutation, or that the mutation was lost by contamination prior to sharing the stock with other labs. Additional *in vitro* experiments performed in the framework of *ReproSci* reveal that PGRP-SD does not contribute to Toll activation by peptidoglycan under experimental conditions (**Supplementary S1**). This suggests that the original results assigning PGRP-SD to the Toll pathway may be due to a genetic background effect.

PGRP-LE is exclusively intracellular and does not participate in melanization

(Takehana et al., 2004 [DOI](#), 2002 [DOI](#)) found that intracellular PGRP-LE contributes to the activation of the Imd pathway together with the transmembrane receptor PGRP-LC. While its role as an intracellular sensor has been confirmed (Bosco-Drayon et al., 2012 [DOI](#); Neyen et al., 2012 [DOI](#)), the initial papers claimed that an extracellular form of PGRP-LE could function akin to CD14 (Takehana et al., 2004 [DOI](#), 2002 [DOI](#)). Follow-up papers have not addressed the existence of this extracellular form. Using a new proteomic approach, we did not detect PGRP-LE in adult fly hemolymph (Rommelaere et al., 2024 [DOI](#)). Moreover, use of a GFP tagged version of PGRP-LE (Bosco-Drayon et al., 2012 [DOI](#)) did not reveal an extracellular form in larvae (**Supplementary S8**). We conclude that PGRP-LE is an exclusively intracellular sensor, with a prominent role regulating the gut immune response (Bosco-Drayon et al., 2012 [DOI](#); Neyen et al., 2012 [DOI](#)). In addition, experiments performed for *ReproSci* found no role for PGRP-LE in melanization, contrary to (Takehana et al., 2004 [DOI](#), 2002 [DOI](#)) (**Supplement S9**). The results of the initial papers are likely due to the use of ubiquitous overexpression of *PGRP-LE*, resulting in melanization due to overactivation of the Imd pathway and resulting tissue damage. We were also unable to find a role for PGRP-LE in survival to *Bacillus* infections (**Supplementary S10**) and conclude that the initial result was due to genetic background and comparison to a robust wild-type as the baseline.

LPS does not activate the Imd pathway

In mammals, LPS is a strong inducer of TLR4-NF-κB pathway (Hoshino et al., 1999 [DOI](#)). Studies in *Drosophila* using LPS from SigmaTM initially provided evidence that LPS can stimulate the Imd pathway (Imler et al., 2000 [DOI](#)). However, use of highly purified bacterial cell wall components showed that DAP-type peptidoglycan and not LPS stimulates Imd pathway activity (Kaneko et al., 2004 [DOI](#); Leulier et al., 2003 [DOI](#)). Furthermore, in contrast to an initial claim by Leulier et al., 2003 [DOI](#), monomeric DAP-type peptidoglycan (notably tracheal cytotoxin, TCT) is also a strong elicitor of the Imd pathway, in addition to polymeric peptidoglycan (Kaneko et al., 2004 [DOI](#); Stenbak et al., 2004 [DOI](#)).

Naked DNA is not immunogenic in *Drosophila*

In mammals, free DNA can activate IFN-mediated inflammatory responses through sensors such as cGAS (Decout et al., 2021 [DOI](#)). In 2002, an article reported that DNA is immunogenic in *Drosophila* after observing elevated expression of *Diptericin* genes in *DNase II*^{lo} hypomorphic mutant flies (Mukae et al., 2002 [DOI](#)). However, experiments performed for *ReproSci* using the original *DNase II*^{lo} hypomorph show that elevated *Diptericin* expression in the hypomorph is eliminated by outcrossing of chromosome II, and does not occur in an independent *DNase II* null mutant, indicating that this effect is due to genetic background (**Supplementary S11**).

Furthermore, elevated *Diptericin* expression in *DNase II^{lo}* flies is not observed in germ-free conditions, indicating that it is not due to DNA excess (West, 2018 [↗](#)). Moreover, experiments showed that injection of naked bacterial and eukaryotic DNA does not activate an immune response in wild-type or *DNase II* mutant flies. These results indicate that naked DNA unlikely to be an elicitor for AMP genes in *Drosophila* as originally claimed, although it may activate the recently discovered cGLR-STING pathway (Cai et al., 2022 [↗](#)). The possibility that DNA-associated proteins such as histones are immunogenic also remains to be investigated.

JNK signaling is not directly required for AMP expression

Several studies pointed to a direct role of the JNK pathway in the regulation of AMPs (Delaney et al., 2006 [↗](#); Kallio et al., 2005 [↗](#); Kleino et al., 2005 [↗](#)) while other studies did not identify any major role (Boutros et al., 2002 [↗](#); Kenmoku et al., 2017 [↗](#); Park et al., 2004 [↗](#); Silverman et al., 2003 [↗](#)). Experiments performed for *ReproSci* and review of the literature suggest that JNK is unlikely to have a strong direct or systematic role on regulation of *Drosophila* AMPs *in vivo*, and is not required in the adult fat body for AMP expression in response to infection (Supplementary S12). Previous results finding a direct role for JNK in AMP production are likely due to a combination of i) use of cell culture systems that do not fully reflect *in vivo* systems; ii) the incorrect conclusion that kinase-dead JNK kinase (TAK1) is biologically equivalent to a null mutant; and iii) underappreciation for the overarching role that JNK plays in cytoskeletal remodeling, pathway activation, and cell differentiation, which are not specific to immunity.

Caspar is not a canonical negative regulator of Imd signaling

caspar encodes an orthologue of human Fas associated factor 1 (FAF1) which was reported to interact with key immune proteins including NF-κB. Kim et al., (2006) found that a null mutation of *caspar* produced high constitutive expression of *Diptericin* and increased resistance to Gram-negative infections. Using flies deficient for *caspar*, we did not detect any increased *Diptericin* expression in unchallenged flies as initially reported (Supplementary S13). Moreover, Caspar overexpression did not result in significantly lower levels of *Diptericin*, and did not promote survival to Gram-negative infections. We conclude that while Caspar might affect the Imd pathway in certain tissue-specific contexts, it is unlikely to act as a generic negative regulator of the Imd pathway.

Insights from the reproducibility project on immune-induced peptides

Edin does not significantly contribute to lifespan or resistance against *L. monocytogenes*

Gordon et al., (2008 [↗](#)) identified *edin*, a host defense peptide gene highly expressed upon *L. monocytogenes* infection. They found that overexpression of *edin* markedly reduced lifespan and fly fitness whereas RNAi knockdown of *edin* in the fat body increased susceptibility to *L. monocytogenes* infection. Contrary to (Gordon et al., 2008 [↗](#)), experiments performed for *ReproSci* using a novel *edin* null mutant indicate that loss of *edin* does not increase susceptibility to *L. monocytogenes* infection compared to appropriate controls. Moreover, overexpression of *edin* did not cause any major reduction in lifespan (Supplementary S14).

Listericin overexpression is not protective against *L. monocytogenes* infection

(Goto et al., 2010 [↗](#)) found using transfection and RNAi in S2 cells that the *Listericin* gene encodes a novel antibacterial peptide-like protein whose induction is cooperatively regulated by PGRP-LE and the JAK-STAT pathway. However, a microarray indicates that this gene is regulated by the Imd pathway with a minor impact of Toll (De Gregorio et al., 2002 [↗](#)). Experiments performed for *ReproSci* suggest that *Listericin* expression does not specifically depend on PGRP-LE but rather is induced through both PGRP-LC and -LE as other Imd-regulated AMPs are (Supplementary S15). Silencing STAT92A in the fat body by using an *in vivo* RNAi approach did not confirm a role of JAK/STAT in induction of *Listericin* following infection as claimed by (Goto et al., 2010 [↗](#)).

Overexpression of Listericin using a newly generated construct did not reveal any protective effect against *L. monocytogenes* as found by (Goto et al., 2010 [↗](#)). Thus, Listericin should be considered as a host defense peptide regulated by the Imd pathway whose activity remains to be confirmed.

Insights from the reproducibility project on phagocytosis and encapsulation

Dscam1 does not directly contribute to phagocytosis through opsonization

Drosophila Dscam1 has three immunoglobulin-like domains and encodes thousands of isoforms through alternate splicing. An early paper proposed that secreted Dscam1 isoforms can bind bacteria and enhance phagocytosis, an attractive hypothesis due to its potential to produce multiple ‘recognition’ isoforms akin to an adaptive immune system (Watson et al., 2005 [↗](#)). Dscam1 homologs have independently been implicated in crustacean and mosquito immunity, systems that are not amenable to genetic studies (Dong et al., 2006 [↗](#); Ng et al., 2014 [↗](#); Ng and Kurtz, 2020 [↗](#)). However, no follow-up studies were published on the role of Dscam1 proteins in *Drosophila* immunity. Subsequent studies showed that *Dscam1* splicing is not affected by septic injury (Armitage et al., 2014 [↗](#)), and contrary to a role for secreted Dscam1 isoforms in *Drosophila* opsonization, a proteomic analysis of adult hemolymph did not detect any secreted Dscam1 isoforms (Rommelaere et al., 2024 [↗](#)). Furthermore, all Dscam1 isoforms encode the transmembrane domain. Experiments performed for *ReproSci* did reveal a mild decrease in phagocytosis of *E. coli* and *S. aureus* by hemocytes of *hml>Dscam1-RNAi* larvae: a 30% reduction in comparison to 50% in the initial study (Watson et al., 2005 [↗](#)) (Supplementary S16). Dscam1 plays a key role in neuronal dendrite arborization and affects how neurons influence peripheral hematopoiesis, and RNAi against *Dscam1* in neurons reduces hemocyte numbers (Ouyang et al., 2020 [↗](#)). It is likely that reduced phagocytosis in *Dscam1* depleted animals is due to an effect on hematopoiesis rather than a direct immune effect. In the absence of additional studies, *Dscam1* should not be considered a *Drosophila* immune gene *sensu stricto*.

PGRP-LC is not a phagocytic receptor

(Rämet et al., 2002 [↗](#)) reported a role of the Imd receptor PGRP-LC in phagocytosis of Gram-negative bacteria in an extensive RNAi screen (Rämet et al., 2002 [↗](#)). Subsequent studies did not reveal any critical role of PGRP-LC in bacterial engulfment (Choe et al., 2002 [↗](#); Melcarne et al., 2019a [↗](#)). It is likely that PGRP-LC indirectly (and modestly) affects phagocytosis through its role in the Imd pathway and associated upregulation of phagocytic receptors (Wong et al., 2017 [↗](#)), but this pattern recognition receptor should not be considered a phagocytic receptor.

Croquemort is not a phagocytic engulfment receptor for apoptotic cells or bacteria

The CD36 scavenger member Croquemort was initially identified as a macrophage-specific phagocytic receptor for apoptotic cells. Use of an overlapping set of deficiencies removing *croquemort* revealed that Croquemort was critical for engulfment of apoptotic cells during embryogenesis (Franc, 1999 [↗](#); Franc et al., 1996 [↗](#)). However, using a clean *croquemort* null mutant, experiments performed for *ReproSci* reveal that *croquemort* is not required for removal of apoptotic cells in embryos (Supplementary S17). *croquemort* mutant phagocytes do display a general defect in phagosome maturation and accumulate acidic vacuoles (similar to *draper* mutants) (Guillou et al., 2016 [↗](#); Han et al., 2014 [↗](#)). It is not known why *croquemort* affects phagosome maturation, but could be due to its role in lipid uptake (Woodcock et al., 2015 [↗](#)). The available data indicate that Croquemort is not a phagocytic receptor required for initial uptake but contributes to sustaining phagocytosis by promoting lysosomal degradation. Of note, another CD36 member, Santa Maria, was recently implicated in the phagocytosis of apoptotic cells by glia (Hilu-Dadia et al., 2025 [↗](#)).

dSR-CI is not required for phagocytosis of bacteria

The scavenger receptor dSR-CI was initially identified as a hemocyte-restricted receptor with high affinity for low density lipoprotein (Pearson et al., 1995 [\[1\]](#)). *In vitro* and RNAi experiments suggested a significant role for dSR-CI in phagocytosis of bacteria (Rämet et al., 2001 [\[2\]](#)). Experiments for *ReproSci* using a newly generated *dSR-CI* null mutant did not reveal any major requirement of dSR-CI in phagocytosis *in vivo* (Supplementary S18). It is possible that dSR-CI has a redundant role in phagocytosis with one or more of the three additional dSR-Cs encoded in the *Drosophila* genome (Ulvila et al., 2011 [\[3\]](#)), or may play another role in hemocytes which remained to be determined. We confirm the (Rämet et al., 2001 [\[2\]](#)) observation that loss of *SR-CI* has no apparent effect on the inducible AMP response.

Retinophilin/Undertaker has no role in phagocytosis

Using overlapping deficiencies in embryos and RNAi in S2 cells, (Cuttell et al., 2008a [\[4\]](#)) found that Undertaker (*uta/rtp*), a Juncophilin, was required for phagocytosis of apoptotic cells and bacteria and links Draper-mediated phagocytosis to Ca^{2+} homeostasis. However, expression of Undertaker is restricted to the adult eye (Thurmond et al., 2019 [\[5\]](#)), which argues against a role in phagocytosis by embryonic macrophages. Using an *rtp¹* null allele, we did not observe any effect of Undertaker on phagocytosis of apoptotic cells or bacteria by larval hemocytes (Supplementary S19). The published data are most consistent with a scenario in which RNAi generated off-target knockdown of a protein related to *retinophilin/undertaker*, while Undertaker itself is unlikely to have a role in phagocytosis. In contrast, independent data support the notion put forth by (Cuttell et al., 2008a [\[4\]](#)) that calcium flux and store-operated calcium entry (including all other components mentioned in this paper) are required for phagocytosis and blood cell activation in animals (Etchegaray et al., 2012 [\[6\]](#); Gronski et al., 2009 [\[7\]](#); Moon et al., 2020 [\[8\]](#); Mortimer et al., 2013 [\[9\]](#); Weavers et al., 2016 [\[10\]](#)).

NimC1 has a co-operative role in phagocytosis of Gram-negative bacteria

An *in vivo* RNAi approach suggested that NimC1 is required for the uptake of Gram-negative bacteria (Kurucz et al., 2007 [\[11\]](#)). In contrast, a null mutation in NimC1 did not reproduce this phenotype (Melcarne et al., 2019). However a *NimC1, eater* double mutant revealed that these two Nimrod receptors function co-operatively in phagocytosis of Gram-negative bacteria (Melcarne et al., 2019b [\[12\]](#)). It is possible this disparity relies on a cryptic effect, such as genetic background differences. Although the initial claim that NimC1 alone is essential for phagocytosis of Gram-negative bacteria was not reproducible, the claim that NimC1 is involved in phagocytosis of Gram-negative bacteria is correct.

Eater has a co-operative role in phagocytosis

An initial study pointed to a role for the transmembrane receptor Eater in the phagocytosis of both Gram-negative and Gram-positive bacteria (Kocks et al., 2005 [\[13\]](#)). The authors used an overlapping set of deficiencies to remove Eater *in vivo* and RNAi in S2 cells to reach this conclusion. While nearly all of the claims in the original article have been confirmed, subsequent studies showed that a null mutation of *eater* does not impact phagocytosis of Gram-negative bacteria (Bretscher et al., 2015 [\[14\]](#); Melcarne et al., 2019). Reduced phagocytosis of Gram-negative bacteria was observed only in *NimC1, eater* double mutants, indicating that Eater is involved in this process but not strictly necessary.

Psidin is not required for induction of Defensin

Brennan et al., (2007 [\[15\]](#)) found that Psidin acts non-autonomously in the hemocytes to control induction of *Defensin* in the fat body, and suggested that a cytokine produced by hemocytes is required to activate systemic *Defensin* expression. In contrast to this, experiments performed for *ReproSci* using two independent *psidin* mutants show that *Defensin* expression is similar to wild type in these lines (Supplementary S20). Furthermore, evidence from multiple papers suggests that this result, and other instances where mutations have been found to specifically eliminate *Defensin* expression, is likely due to segregating polymorphisms within *Defensin* that disrupt primer binding in some genetic backgrounds and lead to a false negative result (Supplementary

S20). (Brennan et al., 2007 [↗](#)) further found that Psidin is required for phagolysosomal maturation. Experiments performed for *ReproSci* confirm that phagolysosomal maturation is impaired in independent *psidin* mutants, and furthermore that *psidin* mutant hemocytes are abnormally large and multinucleate (**Supplementary S20**).

Hemese is not essential for encapsulation

(Kurucz et al., 2003 [↗](#)) found that loss of Hemese, a hemocyte-specific transmembrane protein, causes overproduction of lamellocytes in permissive conditions. In contrast to this, using an independent *hemese*^{SK2} null mutant, we find that loss of *hemese* does not strongly affect lamellocyte number or encapsulation (**Supplementary S21**). The function of Hemese awaits further functional characterization.

Insights from the reproducibility project on IRC, Duox and NOS

Immune Regulated Catalase (IRC) is not essential for survival to benign enteric infection

Using a ubiquitous and directed RNAi approach, (Ha et al., 2005b [↗](#)) found that adult flies depleted of the extracellular catalase IRC (immune regulated catalase) displayed high mortality rates after ingestion of foods contaminated with normally benign microbes (Ha et al., 2005b [↗](#)). Contrary to this, we did not observe any increase in mortality upon feeding IRC null mutant flies with various microbial pathogens, although an effect of the mutation on motor function and overall fitness was observed (**Supplementary S22**). The immune regulated gene *IRC* is likely to play a role in the immune response that remains to be characterized.

Duox-mediated ROS production is most likely a signalling mechanism rather than an antibacterial effector

Early experiments showed that ingestion of pathogenic bacteria by adult flies resulted in rapid upregulation of *Duox* expression in the midgut and an associated *Duox*-dependent reactive oxygen species (ROS) burst as shown by ubiquitous knockdown of this gene. Ha et al., (2005b) [↗](#) inferred that ROS produced by Duox, particularly HOCl, had highly bactericidal effects, as clearance of ingested bacteria was impaired in *Duox-RNAi* flies (Ha et al., 2005a [↗](#)). A role of Duox in the control of the gut microbiota has also been subsequently observed in other insects (Yao et al., 2016 [↗](#)). Epithelium damage associated with Duox-mediated production of ROS was also thought to explain the increased epithelium renewal observed upon oral bacterial infection (Buchon et al., 2009a [↗](#)). However, most of these experiments were performed using a ubiquitous Gal4 driver to silence *Duox*. *Duox* is weakly expressed in the midgut compared to its expression in the adult crop, or larval hindgut and trachea (Thurmond et al., 2019 [↗](#)), and is an essential gene required for development (Hurd et al., 2015 [↗](#)). Experiments done in the framework of *ReproSci* were unable to reproduce a defect in bacterial clearance (*Ecc15-GFP*) in *Duox-RNAi* flies at the same time point as the initial publication (**Supplementary S23**). Furthermore, we did not observe any increase in *Duox* expression in the gut upon oral bacterial infection. Recent studies show that rather than Duox, the alternate *Drosophila* NADPH oxidase Nox stimulates stem cell proliferation upon bacterial infection (Iatsenko et al., 2018 [↗](#); Jones et al., 2013 [↗](#); Patel et al., 2019 [↗](#)). Additionally, a recent article suggests that the impact of Duox on gut epithelial renewal is an indirect effect of Duox in the Malpighian tubules rather than the midgut (Liu et al., 2024 [↗](#)). Other evidence suggests that luminal ROS produced by Duox contributes to an early bacterial clearance via a signaling role that promotes visceral muscle contraction, regardless of its potential bacteriostatic activity (Benguettat et al., 2018 [↗](#); Du et al., 2016 [↗](#)). We conclude that data on a bactericidal role for *Duox* in the gut context remain contradictory. More precise experiments are needed to clearly define the role of *Duox* in gut epithelial immunity, which may primarily play a signaling role rather than a microbicidal one.

Nitric oxide synthase (NOS) is not essential for systemic AMP expression upon oral infection

A study suggested that inhibition of nitric oxide synthase (NOS) by the inhibitor L-NAME increased larval sensitivity to Gram-negative bacterial infection, and reduced expression of a *Diptericin-LacZ* reporter in the fat body (Foley and O'Farrell, 2003 [\[1\]](#)). This study reports that the *NOS* gene was upregulated following infection in hemocytes, and that feeding with an NO donor induced systemic *Diptericin* expression, pointing to a role for NOS activity in mediating a robust innate immune response to Gram-negative bacteria. However, microarrays monitoring the immune response of larvae or adult flies to systemic or oral infections show that NOS is not an immune inducible gene (Buchon et al., 2009; De Gregorio et al., 2001; Vodovar et al., 2005). Few follow-up studies have explored roles of NOS in activation of the Imd pathway (but see (Brown et al., 2009 [\[2\]](#); Chakrabarti et al., 2012 [\[3\]](#); Dijkers and O'Farrell, 2007 [\[4\]](#); McGettigan et al., 2005 [\[5\]](#))). Use of a viable null *NOS* mutant and RT-qPCR did not reveal a striking role for NOS in expression of the *Diptericin* transcript upon natural infection with the Gram-negative bacterium *Ecc15* (Supplementary S24). Loss of NOS may have a subtle impact on reporter expression by affecting fat body maturation, consistent with a role for NOS in regulation of the developmental hormone ecdysone (Jaszczak et al., 2015 [\[6\]](#)). The role of NOS in the gut to fat-body signaling, if any, requires further study.

Insights from the reproducibility project on the roles of Eiger and Gr28b in immunity

Eiger has roles in melanization but minor effects on survival and AMP expression

A series of results implicating the TNF-related gene *eiger* in several aspects of *Drosophila* immune function (Brandt et al., 2004 [\[1\]](#); Mabery and Schneider, 2010 [\[2\]](#); Schneider et al., 2007 [\[3\]](#)) have been called into question due to the recent discovery that widely-used *eiger* mutants bore a secondary null mutation in the gene encoding the phagocytic receptor *NimC1* (Kodra et al., 2020), which has independent roles in immunity (Melcarne et al., 2019). Experiments for *ReproSci* using clean *egr¹* and *egr³* mutants with a wild-type *NimC1* locus (Kodra et al., 2020), and *NimC1¹* mutants for comparison, suggest that both *eiger* and *NimC1* have independent effects on survival to several pathogens, and that the effect of one or both mutations may explain several survival effects previously attributed solely to Eiger. Contrary to previous results, we find that mutation of *eiger* does not increase tolerance to *Salmonella typhimurium* infection, and does not generally increase susceptibility to extracellular pathogens (Supplementary S25, consistent with (Geuking et al., 2009 [\[4\]](#))). However, our results support a role for Eiger in hemolymph melanization in both adult flies and larvae consistent with (Bidla et al., 2007 [\[5\]](#)) (Supplementary S26). Experiments done in the framework of *ReproSci* did not reveal any induction of the *eiger* gene in the fat body, or a role in regulation of antibacterial peptide gene expression as initially described (Supplementary S27). Immune effects of *eiger* mutation are largely consistent with the reduced melanization observed in these mutants.

Mutation of Gr28b does not produce anorexia, impair melanization, or affect humoral immunity

(Ayres and Schneider, 2009 [\[1\]](#)) found that a mutation in the gene encoding the gustatory receptor Gr28b caused constitutive anorexia and affected fly survival to a number of bacteria. Experiments for *ReproSci* instead suggest that the observed immune phenotypes are largely attributable to reduced melanization linked to the genetic background of the original *Gr28b* mutant, as an independent *Gr28b* mutant does not replicate these phenotypes (Supplementary S28). Furthermore, our results and those of (Sang et al., 2019 [\[2\]](#)) indicate that *Gr28b* mutants are not constitutively anorexic, and this result was likely obtained due to comparison to a wild-type line with unusually high feeding rates.

Discussion

With thousands of papers published in the field of *Drosophila* immunity, it is difficult to summarize all we know of the fly immune system (Westlake et al., 2024). Today, the *Drosophila* immune system has been extensively characterized and serves as a model for other insects of medical or agricultural interest. The use of combined genetic approaches makes data obtained in the *Drosophila* model extremely reliable by current scientific standards. However, it is both impossible and undesirable to entirely avoid publication of controversial or irreproducible results, as science is an incremental and exploratory process where progress is made by constant course correction. That being said, contradictory data or claims that are propagated or not clearly refuted can cause a non-negligible loss of time to scientists in research development. Taking into consideration the relevance of *Drosophila* immune research beyond the immediate community and the prestige associated with the use of sophisticated genetic approaches, it is valuable to create easily accessible assessments of established claims in the field, the roots of which may be buried deep in past literature.

In this paper, we have retrospectively analyzed the verifiability of claims from 400 articles accepted for publication before 2011 in the field of *Drosophila* immunity. We restricted our analysis to *Drosophila* host defense against bacteria and fungi, which the host laboratory has expertise in. It would be interesting to extend this analysis to *Drosophila* antiviral and anti-parasitoid immunity with collaboration from relevant laboratories. We expect that application of our approach to viruses and parasitoids would be more challenging, as co-evolved pathogens are expected to display more specific mechanisms that suppress the host immune system (Westlake et al., 2024).

Verifiability of claims was assessed either by analysing subsequent articles or through experimental testing, which we performed for 56 claims. This extensive assessment resource has been centralized on a website that can be accessed (<https://ReproSci.epfl.ch/>). We hope to continue updating this database as currently unchallenged claims are tested. This article summarizes some of the main findings gained through this extensive reproducibility project. We cannot definitively exclude that some claims highlighted here as challenged could in fact be correct, and that our assessment was erroneous. Reproducibility projects are hampered by the fact that verification must be more stringent and convincing than the initial results to be widely accepted (Errington et al., 2021; Goodman et al., 2016; Shiffrin et al., 2018). However, we encourage the community to continue taking care when considering claims that have been challenged by *ReproSci*, which may be further clarified with future studies. In our analysis, we were unable to reach definitive conclusions on the immune roles of IRC, Duox and NOS, although we suggest that their roles are not as simple as proposed in the initial articles.

Except for the articles on the role of 18-wheeler in immunity (Ligoxygakis et al., 2002; Williams et al., 1997), PGRP-SD upstream of Toll (Bischoff et al., 2004; Iatsenko et al., 2016) and stimulation of the immune response by LPS (Imler et al., 2000; Kaneko et al., 2004; Leulier et al., 2003), few published results in the field of *Drosophila* immunity have been formally challenged. While many of the challenged claims discussed in this paper are known to be controversial by experts, or have fallen quietly out of current research interests, some are still a source of confusion. This is particularly true for young scientists new to the field of *Drosophila* immunity or scientists with primary interests outside of *Drosophila* immunity that use the field as a reference.

Where possible, we have attempted to clarify the causes that led to irreproducible data or conclusions, as they emphasize challenges inherent to the field of *Drosophila* immunity and may point to methodologies that require cautious use (Table 1, see Supplementary Table 2 for a list of challenged claims). A first major source of irreproducible results is where the genetic background is responsible for part or all the observed effect rather than the tested mutation. This is often due to use of inappropriate controls, for example where the wild-type comparison has a different genetic background than the mutant of interest. An example of this is the finding that WntD acts in the control of the Toll pathway (Gordon et al., 2005b). Here, we obtained the same

results as the authors of the claim when using the same mutant lines, but the result does not stand when using an independent mutant of the same gene, indicating the result was likely due to genetic background. Hemocyte numbers may also vary strongly according to genetic background and rearing conditions (Ramond et al., 2020 [↗](#); Sorrentino et al., 2004 [↗](#)). This further complicates *in vivo* or *ex vivo* phagocytosis assays, which are often difficult to interpret as the effect of single mutations are often small. In contrast to developmental processes which are more tightly constrained, immune parameters are known to vary strongly among wild-type *Drosophila* lines (Lazzaro et al., 2004 [↗](#)), making analysis of subtle effects difficult. This issue can be addressed in several ways: by using isogenized fly stocks with a controlled genetic background (Ryckebusch et al., 2025 [↗](#)), by testing the phenotype of a mutation in an alternate background (e.g., over a deficiency) to see if it holds true, by validating loss of function phenotypes with RNAi, and by performing rescue experiments. None of these various ways offer a full guarantee, and caution in interpretation of the results is always required. A range of wild-type lines (Oregon-R, Canton S, etc.) can be tested to evaluate the natural variability of a phenotype and avoid one-to-one comparisons between a single wild-type and mutant that may be misleading. In some cases, natural polymorphisms affecting immune genes can cause apparent effects on gene expression by affecting primer binding, as suggested here for Defensin. Indeed, during this research we found that a few of published primers (Neyen et al. 2014 [↗](#)) were not suited for testing AMP expression across genetic backgrounds.

A second source of irreproducible claims is linked to the use of *in vitro* techniques that do not fully reflect results obtained using *in vivo* genetics approaches. An example is the initial implication of GGBP1 in the sensing of LPS; although this remains an important discovery, subsequent studies revealed that the role of GGBP1 was indirect, acting as an adapter between PGRP-SA and ModSP. Overexpression of genes can produce ectopic or excessive phenotypes (e.g. indicating a role for PGRP-LE in melanization) that are not supported using loss-of function mutations.

A third source of irreproducible claims is linked to the use of RNAi, which had many off-target effects when the technique was first introduced in the early 2000s (Seinen et al., 2011 [↗](#)). The claims related to the serine proteases Spheroid, Sphinx and Spirit (Kambris et al., 2006b [↗](#)) and Undertaker (Cuttell et al., 2008b [↗](#)) appear to be examples of this. Newer generations of RNAi lines have greater specificity towards their target genes. Furthermore, use of several independent RNAi constructs for comparison, and inclusion of driver-only and UAS-only controls, helps to avoid off-target and background effects.

A fourth source of irreproducibility is the use of contaminated or nonspecific reagents. The use of Sigma *E. coli* LPS solution as an immune elicitor has been a source of confusion in innate immune studies in both vertebrates and *Drosophila*, as effects attributed to LPS were found to be due to peptidoglycans or lipopeptides also present in the solution (Kaneko et al., 2004 [↗](#)). Immune roles may also be attributed to genes that affect the development of immune organs, as exemplified by mutations of *18-wheeler* that delay fat body maturation (Ligoxygakis et al., 2002 [↗](#)). Many mutations can have similar indirect effects on the immune system, as exemplified by genes involved with ecdysone signaling, which has both indirect and direct effects on immunity that are difficult to disentangle.

Irreproducibility could also be due to differences in rearing conditions, such as flipping regimens that impact the microbiota, dietary composition (Lesperance and Broderick, 2021 [↗](#); Sannino and Dobson, 2023 [↗](#)), and use of antifungal agents in food. Many fly stocks harbor cryptic endosymbionts such as *Wolbachia* or viruses such as Nora virus with subtle phenotypes that can regardless affect immune parameters under certain conditions (Hanson and Lemaitre, 2023 [↗](#); Teixeira et al., 2008 [↗](#)). Nora virus, which can be present in up to 50% of laboratory fly lines, can significantly reduce lifespan and skew the outcomes of long-term survival studies (Franchet et al., 2025 [↗](#); Habayeb, 2006 [↗](#); Hanson and Lemaitre, 2023 [↗](#)). The very high inducibility of AMP genes used to gauge intensity of the immune response can also be a source of confusion. These genes can be induced by 100-1000-fold upon infection, and their fold change is highly dependent on the basal expression level. If basal AMP expression is elevated in a subset of lines due to cryptic infection or differences in microbiota due to flipping frequency or geographic location, apparent differences in

Genetic background effect	39%
In vitro effect	31%
Pressure to emphasize similarities with human immunity	22%
RNAi off-target	20%
Contaminated reagents or strains	12%
Over-interpretation of results ¹	8%
Developmental/maturation effect ²	8%
Excessive effect of overexpression	4%
Complex effect (microbiota, temperature, cryptic infection)	2%

Table 1. Statistics on methods associated with challenged claims

Where possible, the reasons behind major claims that were challenged were categorized (see full list in Supplementary Table S2). This table shows the likely reasons for major claims to be challenged, and the percentage of claims that were challenged due to that reason. Only one major claim per article was considered, as the same methodological issue often applied to multiple claims from the same article (Nb of articles=49). 1: The data were interpreted in a certain (erroneous) way due to a lack of available context at the time that the article was published, or alternative explanations were not effectively ruled out. 2: The effects observed were not due to a direct effect on immunity but instead due to effects on development of immune tissues or organs, including hemocytes.

induction may simply result from variations in their baseline expression. The variable and high inducibility of AMP genes explains why they are often identified in microarray studies. This can however create a challenge when analyzing gene expression level, as a 5-10 fold induction is minimal for many AMPs. AMP induction can be contextualized by adding unchallenged animals as a negative control and bacteria-challenged animals as a positive control, which cover the range of induction of AMP genes. Selection of appropriate gene readouts and standards are also important (Neyen et al., 2014 [↗](#); Troha and Buchon, 2019 [↗](#)). Finally, we observed that a number of challenged claims involve statements that draw parallels with mammalian immune systems. The tendency for granting agencies and journal editors to value *Drosophila* research only as a model for mammalian immunity insidiously pressures *Drosophila* studies to be framed in a human-centric way.

In this article we have not discussed a number of claims, that had no clear follow-up studies and remain unchallenged (see Supplementary Table 3 for a list of unchallenged claims). We encourage members of the *Drosophila* community to test these, or to contribute unpublished data that addresses these claims. Some examples of these include: the regulation of the Toll pathway by the p62 homolog *Drosophila* Ref(2)P (Avila et al., 2002 [↗](#)), the regulation of AMPs independent of Toll and Imd pathways by the transcription factor FOXO (Becker et al., 2010 [↗](#)), the biological significance of Imd pathway activation through cleavage of PGRP-LC by bacterial proteases (Schmidt et al., 2008 [↗](#)), or the role of PGRP-LE in antimicrobial autophagy against intracellular bacteria (*L. monocytogenes*, *Mycobacterium marinum*) (Yano et al., 2008 [↗](#)). In these studies and others, further research should clarify if the observed effect can be reproduced, and if the immune effects are direct or indirect.

In conclusion, our study has assessed the reproducibility of many claims published in the field of *Drosophila* immunity before 2011. Many articles published in this period were simpler in terms of the number of experiments and techniques used than modern studies, allowing for more efficient assessment. Assessing recent articles might be more useful to the community, to reduce the use of irreproducible data in current areas of research. However, this would likely be challenging as we would lose the benefit of hindsight and progress in the field that comes with time. Here we have favored analysis by claim rather than by article, which allows greater nuance and illustrates that articles may contain both reproducible and irreproducible claims. This highlights the fact that many articles have scientific merit, despite containing challenged claims.

One important goal of this project was to make assessments of claims mainly debated in closed circles more widely accessible, and prevent time loss for scientists who may base research on claims that are widely (but quietly) considered irreproducible. We hope that this effort can assist our community by focusing attention on more promising areas of research, by providing a retrospective on foundational findings in the field, and by identifying common sources of irreproducibility that can be controlled for in the future. In the framework of this project, we opened a web interface that allows the community to comment on these claims and contribute to their assessment. The program used to generate the database is open access, and could eventually be used by other communities to centralize post-publication assessments of articles by claims across research fields.

While we recognize that our approach has subjective elements (identification of claims, choice of claims to test experimentally), we hope that the community-accessible website will allow researchers to voice their opinions and contribute data from many sources to improve objectivity. Our intent is to honor the efforts of the *Drosophila* community while providing a complement that combats the bias against publishing negative results, which can regardless be essential and informative. A companion article (Lemaître et al., 2025 [↗](#)) discusses the metascience aspect of this project by analyzing patterns of irreproducibility and digging into some sociological reasons behind the publication of erroneous data.

Methods

Generation of the list of annotated articles

The purpose of the *ReproSci* reproducibility project is to determine whether a wide range of claims published in the field of *Drosophila* immunity are verifiable using independent sources. A list of publications was generated using a curated search string on the publicly available PubMed database. This search string (**Box1**) included keywords, MeSH terms and prominent authors in the *Drosophila* immunity field, and generated an initial list of approximately 2000 publications. This pool was then manually curated to remove 1) reviews and opinion pieces, 2) topics unrelated to *Drosophila* immunity captured by the search string, and 3) immune topics outside the scope of Lemaitre lab expertise, including development, cell biology, immunity to viruses, immunity to parasitoid wasps, and immunity in other species. Select articles not captured by the search string but known and considered to be key articles in the field were manually added to this selection. The list was then restricted to the date range of interest (1940-2011). This limitation was imposed to allow a time frame in which follow-up studies may have been published that may verify, expand on or contradict the content of the publications selected for the study. These restrictions resulted in a primary list of 400 articles.

Box 1. Search string used on PubMed to generate the initial pool of publications that was subsequently manually curated

The list of articles delivered by the first search string, those that were removed and added, and the final selected list can be found in Table S1. (((“*Drosophila* proteins”[mesh] and (“antimicrobial cationic peptides”[mesh] or immunity)) or “*Drosophila melanogaster*/immunology”[MeSH Terms] or “*Drosophila*/immunology”[MeSH Terms] or (((dimarcq, jean luc[author] or (meister, marie[author] or (reichhart, jean marc[author] or (ligoxigakis, petros[author] or (lemaitre, bruno[author] or (leulier, francois[author] or (kuraishi, takayuki[author] or (royet, julien[author] or (carton, yves[author] or (hoffmann, ja[author] or (ferrandon, dominique[author] or (immler, jean luc[author] or (lagueux, marie[author] or (ezekowitz, r alan[author] or (franc, nathalie c[author] or (maniatas, tom[author] or (silverman n[author] or (perrimon, norbert[author] or (agaisse, herve[author] or (cherry, sara[author] or (medzhitov, ruslan[author] or (engstrom, ylva[author] or (boman, hans[author] or (faye, ingrid[author] or (steiner, hakan[author] or (hultmark d[author] or (dushay, mitchell[author] or (theopold, ulrich[author] or (hedengren m[author] or (kimbrell, deborah a[author] or (anderson k[author] or (schneider, david[author] or (dionne, marc[author] or (wu, louisa[author] or (steward, ruth[author] or (govind, shubha[author] or (clark, andrew[author] or (schlenke, todd[author] or (lazzaro, brian[author] or (unckless, robert[author] or (wasserman s[author] or (eldon, elizabeth[author] or (williams mj[author] or (kurata, shoichiro[author] or (banerjee u[author] or (rizki[author] or (ashburner m[author] or (kimbrell da[author] or (teixeira l[author] or (jiggins, frank[author] or (Agaisse H[Author] or (Boutros M[Author] or (Rahme LG[Author] or (Steward R[Author] or (Lee WJ[Author] or (Nappi A[Author] or (Carton Y[Author] or (Steiner H[Author] or (Jiggins F[Author] or (Poirié M[Author] or (Fauvarque MO[Author] or (Kraaijeveld K[Author] or (Godfray HC[Author] or (Andó I[Author] or (Watnick PI[Author] or (Söderhäll K[Author] or (Chung J[Author] or (Eleftherianos I[Author] or (Roeder T[Author] or (Saleh MC[Author] or (Andino R[Author] or (Kubli E[Author] or (van Rij RP[Author] or (Kawabata SI[Author] or (Ip YT[Author] or (Levine M[Author] or (Ding SW[Author] or (Brennan CA[Author] or (Rämet M[Author] or (Wood W[Author] or (Foley E[Author] or (Gateff E[Author] or (Fauvarque MO[Author] or (O’Farrell PH[Author] or (Matova N[Author] or (Prévost G[Author] or (Schoofs L[Author] or (Pastor-Pareja JC[Author] or (Contamine D[Author] or (Pan D[Author] or (Crozatier M[author] or (Galko MJ[author] or (Ayres J[Author])) and *Drosophila*))

Article annotation

Selected primary articles were annotated by a single researcher (HW) and reviewed by a second researcher (BL) to ensure consistency across the project. The *ReproSci* online database (<https://ReproSci.epfl.ch/>) was constructed to organize the annotations and provide an interactive and searchable interface. Claims were extracted through careful reading of the selected primary articles and organized according to their importance within the paper: a single **Main claim** representing the title claim of the paper or overarching finding; 2-4 **Major claims** emphasized in the abstract or those contributing to the overarching finding; and 0-4 **Minor claims** representing other findings or note not emphasized in the abstract. Wording of the claims was paraphrased and structured in an attempt to represent claims made directly by the authors, rather than interpretations made by the reader. Other supporting data such as the methods used in the paper and evidence used to support claims was also summarized.

Claims were then cross-checked with evidence from previous, contemporary and subsequent publications and assigned a verification category (see [Box 2](#)). A short statement explaining the evidence and citing sources was written to justify the categorization and provide additional context for the claim; this assessment was linked to each claim and can be viewed on the *ReproSci* website. Supplementary documents summarizing complex or ambiguous topics were also written where necessary and can be viewed on the *ReproSci* website.

Box 2. Verification categories used to assess claims

Category	Characteristics
<i>Verified</i>	An independent lab or lab(s) have generated results that directly support this claim using similar or alternate methods.
<i>Verified by same authors</i>	Subsequent work by the same lab or authors supports or extends this claim, but independent verification is not available.
<i>Verified by reproducibility project</i>	- This claim is supported by experiments executed specifically to assess it in the Lemaitre Lab or other laboratories in collaboration with the <i>ReproSci</i> project. - A link to a writeup of the experimental results is included.
<i>Partially verified</i>	- Part of the claim is verified by additional data, but part of it is contradicted. OR - The claim is not supported by subsequent research, but the data used to justify the claim are consistent with current knowledge. The data were interpreted incorrectly either due to lack of context at the time or other confounding factors. OR - The claim is supported by subsequent research, but the data used to justify the claim in the original publication are not
<i>Unchallenged</i>	No additional data is available that significantly supports or contradicts this claim.
<i>Unchallenged, logically consistent</i>	No additional data is available that directly supports this claim, but the claim is consistent with other well-supported related data in the field. .
<i>Unchallenged, logically inconsistent</i>	No additional data is available that directly contradicts this claim, but the claim is inconsistent with other well-supported related data in the field.
<i>Mixed</i>	Both data that supports and contradicts this claim has been published in the field. It is unclear based on the published data whether the claim is correct, incorrect, or whether our understanding of the topic is simply incomplete.
<i>Challenged</i>	An independent lab or lab(s) have generated results that directly contradict this claim using similar or alternate methods.
<i>Challenged by same authors</i>	Subsequent work by the same lab or authors contradicts this claim.
<i>Challenged by reproducibility project</i>	- This claim is contradicted by experiments executed specifically to assess it in the Lemaitre Lab or other laboratories in collaboration with the <i>ReproSci</i> project. - A link to a writeup of the experimental results is included.

Experimental validation

A subset of claims that lacked follow-up in the published literature were selected for direct experimental reproduction by various authors from different laboratory with relevant expertise. This project is concerned with the accuracy of the conclusions themselves rather than the reliability of methods, so we employed indirect or inferential reproducibility, where experimental procedures that may be different from those originally used to reach a conclusion are used to independently verify it. When completed, a document was generated detailing the experiments, including methods and conclusions, conducted to verify an individual claim, which here are summarized as supplementary information. HW and BL take responsibility for the statements made following experimental validation of claims, and acknowledge that our verification experiments could also be erroneous.

These short research projects were also uploaded to the *ReproSci* website, and the assessment of the claim was adjusted from ‘Unchallenged’ to ‘Verified/Challenged by reproducibility project’ as appropriate. The *ReproSci* website was made available to the *Drosophila* scientific community on July 11th 2023, with an email to encourage community members to comment on annotations and verifications and contribute evidence. Experiments performed by various laboratories in the

framework of the *ReproSci* project are summarized in supplemental documents S1 to S28 with their own materials and methods. Additional experimental validation of claims not highlighted in this article can be found on the *ReproSci* website.

The *ReproSci* web interface

A versatile web interface was generated to display the 400 articles with annotations and assessments of the reproducibility of the claims made in these articles. This web interface allows i) easy generation of a defined database of articles including titles, authors, date, abstract; ii) an interface for annotation of each article in terms of major, main and minor claims, as well as fields for methods and additional supporting information, iii) a system that allows each claim or individual evidence to be assessed as verified, unchallenged, challenged, with an associated written assessment including references that were used to assign the claims to a chosen category, iv) outputs that tabulate and enumerate articles and claims according to assessment categories in the database, providing data that allow analysis of multiple parameters of reproducibility, and v) an interactive interface allowing community members to comment on and discuss claims and assessments, including uploading of documents to support their discussion. The implementation of the web interface involved the creation of a Ruby-on-Rails web application associated with a PostgreSQL database to store the data in a structured manner. At least 2 main tables store on one hand the annotations and on the other hand the relationships between them (assessments, evidence, comments). Annotations were indexed on-the-fly with the SOLR search engine, allowing subsequent free text search on them. A large part of Javascript code runs on the client side to assist users while typing annotations. Importantly, the application allows registered users to create new workspaces, each workspace focusing on a research subject. In a workspace, the owner of the project is able to moderate and to assign roles to certain users, such as the ability to add articles, enter annotations and assessments, or comment. Through moderation, community comments may be considered, and assessments updated with new information. The code of this web interface can be found at <https://github.com/fabdavid/ReproSci/> .

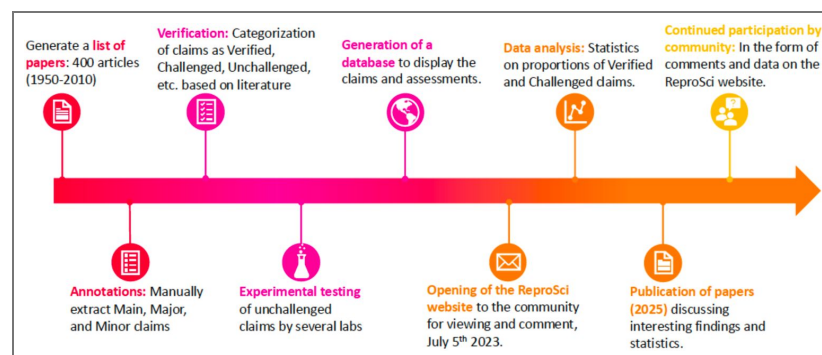


Figure 1. The timeline of the *Drosophila* immunity ReproSci project.

Acknowledgements

We thank the Bloomington Stock Center in the USA and the Vienna *Drosophila* Resource Center (VDRC) for fly stocks; Paola Bellosta, Oren Schuldiner, Julien Royet, Michèle Crozatier and many members of the *Drosophila* community for providing fly strains or wasps; the BioImaging & Optics Platform (BIOP) in EPFL for confocal microscopy and the BioInformatics Competence Center of UNIL-EPFL for data analysis. We thank Neal Silverman for sharing information on Dnase II. This work was supported by Swiss National Science Foundation 310030_189085 (BL) and the open-source program by ETH-Domain's Open Research Data (ORD) Program (2022).

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Funding

Funder	Grant reference number
Swiss National Science Foundation	310030_189085
ETH-Domain's Open Research Data Program 2022	

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Additional files

[Supplemental information S1-S28](#) 

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Peer reviews

Reviewer #1 (Public review):

Summary:

This work revisits a substantial part of the published literature in the field of *Drosophila* innate immunity from 1959 to 2011. The strategy has been to restrain the analysis to some 400 articles and then to extract a main claim, two to four major claims and up to four minor claims totaling some 2000 claims overall. The consistency of these claims with the current state-of-the-art has been evaluated and reported on a dedicated Web site known as ReproSci and also in the text as well as in the 28 Supplements that report experimental verification, direct or indirect, e.g., using novel null mutants unavailable at the time, of a selected set of claims made in several articles. Of note, this review is mostly limited to the manuscript and its associated supplements and does not integrally cover the ReproSci website.

Strengths:

One major strength of this article is that it tackles the issue of reproducibility/consistency on a large scale. Indeed, while many investigators have some serious doubts about some results found in the literature, few have the courage, or the means and time, to seriously challenge studies, especially if published by leaders in the field. The Discussion adequately states the major limitations of the ReproSci approach, which should be kept in mind by the reader to form their own opinion.

This study also allows investigators not familiar with the field to have a clearer understanding of the questions at stake and to derive a more coherent global picture that allows them to better frame their own scientific questions. Besides a thorough and up-to-date knowledge of the literature used to assess the consistency of the claims with our current knowledge, a merit of this study is the undertaking of independent experiments to address some puzzling findings and the evidence presented is often convincing, albeit one should keep in mind the inherent limitations as several parameters are difficult to control, especially in the field of infections, as underlined by the authors themselves. Importantly, some work of the lead author has also been re-evaluated (Supplements S2-S4). Thus, while utmost caution should be exerted, and often is, in challenging claims, even if the challenge eventually proves to be not grounded, it is valuable to point out potential controversial issues to the scientific community.

While this is not a point of this review, it should be acknowledged that the possibility to post comments on the ReproSci website will allow further readjustments by the community in the appreciation of the literature and also of the ReproSci assessments themselves and of its complementary additional experiments.

Weaknesses:

Challenging the results from articles is, by its very nature, a highly sensitive issue, and utmost care should be taken when challenging claims. While the authors generally acknowledge the limitations of their approach in the main text and Supplements, there are a few instances where their challenges remain questionable and should be reassessed. This is certainly the

case for Supplement S18, for which the ReproSci authors make a claim for a point that was not made in the publication under scrutiny. The authors of that study (Ramet et al., Immunity, 2001) never claimed that scavenger receptor SR-CI is a phagocytosis receptor, but that it is required for optimal binding of S2 cells to bacteria. Westlake et al. here have tested for a role of this scavenger receptor in phagocytosis, which had not been tested by Ramet et al. Thus, even though the ReproSci study brings additional knowledge to our understanding of the function of SR-CI by directly testing its involvement in phagocytosis by larval hemocytes, it did not address the major point of the Ramet et al. study, SR-CI binding to bacteria, and thus inappropriately concludes in Supplement S18 that "Contrary to (Ramet et al., 2001, Saleh et al., 2006), we find that SR-CI is unlikely to be a major *Drosophila* phagocytic receptor for bacteria *in vivo*." It follows that the results of Ramet et al. cannot be challenged by ReproSci as it did not address this program. Of note, Saleh et al. (2006) also mistakenly stated that SR-CI impaired phagocytosis in S2 cells and could be used as a positive control to monitor phagocytosis in S2 cells. Their assay appears to have actually not monitored phagocytosis but the association of FITC-labeled bacteria to S2 cells by FACS, as they did not mention quenching the fluorescence of bacteria associated with the surface with Trypan blue.

The inference method to assess the consistency of results with current knowledge also has limitations that should be better acknowledged. At times, the argument is made that the gene under scrutiny may not be expressed at the right time according to large-scale data or that the gene product was not detected in the hemolymph by a mass-spectrometry approach. While being in theory strong arguments, some genes, for instance, those encoding proteases at the apex of proteolytic activation cascades, need not necessarily be strongly expressed and might be released by a few cells. In addition, we are often lacking relevant information on the expression of genes of interest upon specific immune challenges such as infections with such and such pathogens.

As regards mass spectrometry, there is always the issue of sensitivity that limits the force of the argument. Our understanding of melanization remains currently limited, and methods are lacking to accurately measure the killing activity associated with the triggering of the proPO activation cascade. In this study, the authors monitor only the blackening reaction of the wound site based on a semi-quantitative measurement. They are not attempting to use other assays, such as monitoring the cleavage of proPOs into active POs or measuring PO enzymatic activity. These techniques are sometimes difficult to implement, and they suffer at times from variability. Thus, caution should be exerted when drawing conclusions from just monitoring the melanization of wounds.

Likewise, the study of phagocytosis is limited by several factors. As most studies in the field focus on adults, the potential role of phagocytosis in controlling Gram-negative bacterial infections is often masked by the efficiency of the strong IMD-mediated systemic immune response mediated by AMPs (Hanson et al, eLife, 2019). This problem can be bypassed in rare instances of intestinal infections by Gram-negative bacteria such as *Serratia marcescens* (Nehme et al., PLoS Pathogens, 2007) or *Pseudomonas aeruginosa* (Limmer et al. PNAS, 2011), which escape from the digestive tract into the hemocoel without triggering, at least initially, the systemic immune response. It is technically feasible to monitor bacterial uptake in adults by injecting fluorescently labeled bacteria and subsequently quenching the signal from non-ingested bacteria. Nonetheless, many investigators prefer to resort to *ex vivo* assays starting from hemocytes collected from third-instar wandering larvae as they are easier to collect and then to analyze, e.g., by FACS. However, it should be pointed out that these hemocytes have been strongly exposed to a peak of ecdysone, which may alter their properties. Like for S2 cells, it is thus not clear whether third-instar larval hemocytes faithfully reproduce the situation in adults. The phagocytic assays are often performed with killed bacteria. Evidence with live microorganisms is better, especially with pathogens. Assays with live bacteria require however, an antibody used in a differential permeabilization protocol. Furthermore, the killing method alters the surface of the microorganisms, a key property for phagocytic

uptake. Bacterial surface changes are minimal when microorganisms are killed by X-ray or UV light. These limitations should be kept in mind when proceeding to inference analysis of the consistency of claims. Eater illustrates this point well. Westlake et al. state that: "[...] subsequent studies showed that a null mutation of eater does not impact phagocytosis". The authors refer here to Bretscher et al., *Biology Open*, 2015, in which binding to heat-killed *E. coli* was assessed in an ex vivo assay in third instar larvae. In contrast, Chung and Kocks (JBC, 2011) tested whether the recombinant extracellular N-terminal ligand-binding domain was able to bind to bacteria. They found that this domain binds to live Gram-positive bacteria but not to live Gram-negative bacteria. For the latter, killing bacteria with ethanol or heating, but not by formaldehyde treatment, allowed binding. More importantly, Chung and Kocks documented a complex picture in which AMPs may be needed to permeabilize the Gram-negative bacterial cell wall that would then allow access of at least the recombinant secreted Eater extracellular domain to peptidoglycan or peptidoglycan-associated molecules. Thus, the systemic Imd-dependent immune response would be required in vivo to allow Eater-dependent uptake of Gram-negative bacteria by adult hemocytes. In ex vivo assays, any AMPs may be diluted too much to effectively attack the bacterial membrane. A prediction is then that there should be an altered phagocytosis of Gram-negative bacteria in IMD-pathway mutants, e.g., an imd null mutant but not the hypomorphic imd[1] allele. This could easily be tested by ReproSci using the adult phagocytosis assay used by Kocks et al, *Cell*, 2005. At the very least, the part on the role of Eater in phagocytosis should take the Chung & Kocks study into account, and the conclusions modulated.

Another point is that some mutant phenotypes may be highly sensitive to the genetic background, for instance, even after isogenization in two different backgrounds. In the framework of a Reproducibility project, there might be no other option for such cases than direct reproduction of the experiment as relying solely on inference may not be reliable enough.

With respect to the experimental part, some minor weaknesses have been noted. The authors rely on survival to infection experiments, but often do not show any control experiments with mock-challenged or noninfected mutant fly lines. In some cases, monitoring the microbial burden would have strengthened the evidence. For long survival experiments, a check on the health status of the lines (viral microbiota, *Wolbachia*) would have been welcome. Also, the experimental validation of reagents, RNAi lines, or KO lines is not documented in all cases.

<https://doi.org/10.7554/eLife.108404.1.sa3>

Reviewer #2 (Public review):

Summary:

The authors present an ambitious and large-scale reproducibility analysis of 400 articles on *Drosophila* immunity published before 2011. They extract major and minor claims from each article, assess their verifiability through literature comparison and, when possible, through targeted experimental re-testing, and synthesize their findings in an openly accessible online database. The goal is to provide clarity to the community regarding claims that have been contradicted, incompletely supported, or insufficiently followed up in the literature, and to foster broader community participation in evaluating historical findings. The manuscript summarizes the major insights emerging from this systematic effort.

Strengths:

(1) Novelty and community value: This work represents a rare example of a systematic, transparent, and community-facing reproducibility project in a specific research domain. The

creation of a dedicated public platform for disseminating and discussing these assessments is particularly innovative.

(2) Breadth and depth: The authors analyze an impressive number of publications spanning multiple decades, and they couple literature-based assessments with new experimental data where follow-up is missing.

(3) Clarity of purpose: The manuscript carefully distinguishes between assessing evidential support for claims and judging the scientific merit of historical work. This helps frame the project as constructive rather than punitive.

(4) Metascientific relevance: The analysis identifies methodological and contextual factors that commonly underlie irreproducible claims, providing a useful guide for future study design and interpretation.

(5) Transparency: Supplementary datasets and the public website provide an exceptional degree of openness, which should facilitate community engagement and further refinement.

Weaknesses:

(1) Subjectivity in selection: Despite the authors' efforts, the choice of which papers and claims to highlight cannot be entirely objective. This is an inherent limitation of any retrospective curation effort, but it remains important to acknowledge explicitly.

(2) Emphasis on irreproducible claims: The manuscript focuses primarily on claims that are challenged or found to be weakly supported. While understandable from the perspective of novelty, this emphasis may risk overshadowing the value of claims that are well supported and reproducible.

(3) Framing and language: Certain passages could benefit from more neutral phrasing and avoidance of binary terms such as "correct" or "incorrect," in keeping with the open-ended and iterative nature of scientific progress.

(4) Community interaction with the dataset: While the website is an excellent resource, the manuscript could further clarify how the community is expected to contribute, challenge, or refine the annotations, especially given the large volume of supplementary data.

(5) Minor inconsistency: The manuscript states that papers from 1959-2011 were included, but the Methods section mentions a range beginning in 1940. This should be aligned for clarity.

Impact and significance:

This contribution is likely to have a meaningful impact on both the *Drosophila* immunity community and the broader scientific ecosystem. It highlights methodological pitfalls, encourages transparent post-publication evaluation, and offers a reusable framework that other fields could adopt. The work also has pedagogical value for early-career researchers entering the field, who often struggle to navigate contradictory or outdated claims. By centralizing and contextualizing these discussions, the manuscript should help accelerate more robust and reproducible research.

<https://doi.org/10.7554/eLife.108404.1.sa2>

Reviewer #3 (Public review):

Summary:

In this ambitious study, the authors set out to analyse the validity of a number of claims, both minor and major, from 400 published articles within the field of *Drosophila* immunity that were published before 2011. The authors were able to determine initially if claims were supported by comparing them to other published literature in the field and, if required, by experimentally testing 'unchallenged' claims that had not been followed up in subsequent published literature. Using this approach, the authors identified a number of claims that had contradictory evidence using new methods or taking into account developments within the field post-initial publication. They put their findings on a publicly available website designed to enable the research community to assess published work within the field with greater clarity.

Strengths:

The work presented is rigorous and methodical, the data presentation is high quality, and importantly, the data presented support the conclusions. The discussion is balanced, and the study is written considerately and respectfully, highlighting that the aim of the study is not to assign merit to individual scientists or publications but rather to improve clarity for scientists across the field. The approach carried out by the researchers focuses on testing the validity of the claims made in the original papers rather than testing whether the original experimental methods produced reproducible results. This is an important point since there are many reasons why the original interpretation of data may have understandably led to the claims made. These potential explanations for irreproducible data or conclusions are discussed in detail by the authors for each claim investigated.

The authors have generated an accompanying website, which provides a valuable tool for the *Drosophila* Immunity research community that can be used to fact-check key claims and encourages community engagement. This will achieve one important goal of this study - to prevent time loss for scientists who base their research on claims that are irreproducible. The authors rightly point out that it is impossible (and indeed undesirable) to avoid publication of irreproducible results within a field since science is 'an exploratory process where progress is made by constant course correction'. This study is, however, an important piece of work that will make that course correction more efficient.

Weaknesses:

I have little to recommend for the improvement of this manuscript. As outlined in my comments above, I am very supportive of this manuscript and think it is a bold and ambitious body of work that is important for the *Drosophila* immunity field and beyond.

<https://doi.org/10.7554/eLife.108404.1.sa1>

Reviewer #4 (Public review):

This is an important paper that can do much to set an example for thoughtful and rigorous evaluation of a discipline-wide body of literature. The compiled website of publications in *Drosophila* immunity is by itself a valuable contribution to the field. There is much to praise in this work, especially including the extensive and careful evaluation of the published literature. However, there are also cautions.

One notable concern is that the validation experiments are generally done at low sample sizes and low replication rates, and often lack statistical analysis. This is slippery ground for declaring a published study to be untrue. Since the conclusions reported here are nearly all negative, it is essential that the experiments be performed with adequate power to detect the originally described effects. At a minimum, they should be performed with the same sample size and replication structure as the originally reported studies.

The first section of Results should be an overview of the general accuracy of the literature. Of all claims made in the 400 evaluated papers, what proportion fell into each category of "verified", "unchallenged", "challenged", "mixed", or "partially verified"? This summary overview would provide a valuable assessment of the field as a whole. A detailed dispute of individual highlighted claims could follow the summary overview.

Section headings are phrased as declarative statements, "Gene X is not involved in process Y", which is more definitive phrasing than we typically use in scientific research. It implies proving a negative, which is difficult and rare, and the evidence provided in the present manuscript generally does not reach that threshold. A more common phrasing would be "We find no evidence that gene X contributes to process Y". A good model for this more qualified phrasing is the "We conclude that while Caspar might affect the Imd pathway in certain tissue-specific contexts, it is unlikely to act as a generic negative regulator of the Imd pathway," concluding the section on the role of Caspar. I am sure the authors feel that the softer, more qualified phrasing would undermine their article's goal of cleansing the literature of inaccuracies, but the hard declarative 'never' statements are difficult to justify unless every validation experiment is done with a high degree of rigor under a variety of experimental conditions. This caveat is acknowledged in the 3rd paragraph of the Discussion, but it is not reflected in the writing of the Results. The caveat should also appear in the Introduction.

The article is clear that "Claims were assessed as verified, unchallenged, challenged, mixed, or partially verified," but the project is called "reproducibility project" in the 7th line of the abstract, and the website is "ReproSci". The fourth line of the abstract and the introduction call some published research "irreproducible". Most of the present manuscript does not describe reproduction or replication. It describes validation, or independent experimental tests for consistency. Published work is considered validated if subsequent studies using distinct approaches yielded consistent results. For work that the authors consider suspicious, or that has not been subsequently tested, the new experiments provided here do not necessarily recreate the published experiment. Instead, the published result is evaluated with experiments that use different tools or methods, again testing for consistency of results. This is an important form of validation, but it is not reproduction, and it should not be referred to as such. I strongly suggest that variations of the words "reproducible" or "replication" be removed from the manuscript and replaced with "validation". This will be more scientifically accurate and will have the additional benefit of reducing the emotional charge that can be associated with declaring published research to be irreproducible.

The manuscript includes an explanatory passage in the Results section, "Our project focuses on assessing the strength of the claims themselves (inferential/indirect reproducibility) rather than testing whether the original methods produce repeatable results (results/direct reproducibility). Thus, our conclusions do not directly challenge the initial results leading to a claim, but rather the general applicability of the claim itself." Rather than first appearing in Results, this statement should appear prominently in the abstract and introduction because it is a core element of the premise of the study. This can be combined with the content of the present Disclaimer section into a single paragraph in the Introduction instead of appearing in two redundant passages. I would again encourage the authors to substitute the word validation for reproduction, which would eliminate the need for the invented distinction between indirect versus direct reproduction. It is notable that the authors have chosen to title the relevant Methods section "Experimental Validation" and not "Replication".

Experimental data "from various laboratories" in the last paragraph of the Introduction and the first paragraph of the Results are ambiguous. Since these new experiments are part of the central core of the manuscript, the specific laboratories contributing them should be named in the two paragraphs. If experiments are being contributed by all authors on the manuscript, it would suffice to say "the authors' laboratories". The attribution to "various labs" appears to

be contradicted by the Discussion paragraph 2, which states "the host laboratory has expertise in" antibacterial and antifungal defense, implying a single lab. The claim of expertise by the lead author's laboratory is unnecessary and can be deleted if the Lemaitre lab is the ultimate source of all validation experiments.

The passage on the controversial role of Duox in the gut is balanced and scholarly, and stands out for its discussion of multiple alternative lines of evidence in the published literature and supplement. This passage may benefit from research by multiple groups following up on the original claims that are not available for other claims, but the tone of the Duox section can be a model for the other sections.

Comments on other sections and supplements:

I understand the desire to explain how original results may have been obtained when they are not substantiated by subsequent experiments. However, statements such as "The initial results may have been obtained due to residual impurities in preparations of recombinant GGBP1" and "Non-replicable results on the roles of Spirit, Sphinx and Spherioide in Toll pathway activation may be due to off-target effects common to first-generation RNAi tools" are speculation. No experimental data are presented to support these assertions, so these statements and others like them (currently at the end of most "insights" sections) should not appear in Results. I recognize that the authors are trying to soften their criticism of prior studies by providing explanations for how errors may have occurred innocently. If they wish to do so, the speculative hypotheses should appear in the Discussion.

The statement in Results that "The initial claim concerning wntD may be explained by a genetic background effect independent of wntD" similarly appears to be a speculation based on the reading of the main text Results. However, the Discussion clarifies that "Here, we obtained the same results as the authors of the claim when using the same mutant lines, but the result does not stand when using an independent mutant of the same gene, indicating the result was likely due to genetic background." That additional explanation in the Discussion greatly increases reader confidence in the Result and should be explained with reference to S5 in the Results. Such complete explanations should be provided everywhere possible without requiring the reader to check the Supplement in each instance.

In some cases, such as "The results of the initial papers are likely due to the use of ubiquitous overexpression of PGRP-LE, resulting in melanization due to overactivation of the Imd pathway and resulting tissue damage", the claim to explain the original finding would be easy to test. The authors should perform those tests where they can, if they wish to retain the statements in the manuscript. Similarly, the claim "The published data are most consistent with a scenario in which RNAi generated off-target knockdown of a protein related to retinophilin/undertaker, while Undertaker itself is unlikely to have a role in phagocytosis" would be stronger if the authors searched the *Drosophila* genome for a plausible homolog that might have been impacted by the RNAi construct, and then put forth an argument as to why the off-target gene is more likely to have generated the original phenotype than the nominally targeted gene. There is a brief mention in S19 that junctophilin is the authors' preferred off-target candidate, but no evidence or rationale is presented to support that assertion. If the original RNAi line is still available, it would be easy enough to test whether junctophilin is knocked down as an off-target, and ideally then to use an independent knockdown of junctophilin to recapitulate the original phenotype. Otherwise, the off-target knockdown hypothesis is idle speculation.

A good model is the passage on extracellular DNA, which states, "experiments performed for ReproSci using the original DNase II^{lo} hypomorph show that elevated Dipterocin expression in the hypomorph is eliminated by outcrossing of chromosome II, and does not occur in an independent DNase II null mutant, indicating that this effect is due to genetic background (Supplementary S11)." In this case, the authors have performed a clear experiment that

explains the original finding, and inclusion of that explanation is warranted. Similar background replacement experiments in other validations are equally compelling.

The statement "Analysis of several fly stocks expected to carry the PGRP-SDdS3 mutation used in the initial study revealed the presence of a wild-type copy PGRP-SD, suggesting that either the stock used in this study did not carry the expected mutation, or that the mutation was lost by contamination prior to sharing the stock with other labs" provides a documentable explanation of a potential error in the original two manuscripts, but the subsequent "analysis of several fly stocks" needs citations to published literature or explanation in the supplement. It is unclear from this passage how the wildtype allele in the purportedly mutant stocks could have led to the misattribution of function to PGRP-SD, so that should be explained more clearly in the manuscript.

The originally claimed anorexia of the Gr28b mutation is explained as having been "likely obtained due to comparison to a wild-type line with unusually high feeding rates". This claim would be stronger if the wildtype line in question were named and data showing a high rate of feeding were presented in the supplement or cited from published literature. Otherwise, this appears to be speculation.

In the section "The Toll immune pathway is not negatively regulated by wntD", FlyAtlas is cited as evidence that wntD is not expressed in adult flies. However, the FlyAtlas data is not adequately sensitive to make this claim conclusively. If the present authors wish to state that wntD is not expressed in adults, they should do a thorough test themselves and report it in the Supplement.

Alternatively, the statement "data from FlyAtlas show that wntD is only expressed at the embryonic stage and not at the adult stage at which the experiments were performed by (Gordon et al., 2005a)" could be rephrased to something like "data from FlyAtlas show strong expression of wntD in the embryo but not the adult" and it should be followed by a direct statement that adult expression was also found to be near-undetectable by qPCR in supplement S5. That data is currently "not shown" in the supplement, but it should be shown because this is a central result that is being used to refute the original claim. This manuscript passage should also describe the expression data described in Gordon et al. (2005), for contrast, which was an experimental demonstration of expression in the embryo and a claim "RT-PCR was used to confirm expression of endogenous wntD RNA in adults (data not shown)."

Inclusion of the section on croquemort is curious because it seems to be focused exclusively on clearance of apoptotic cells in the embryo, not on anything related to immunity. The subsection is titled "Croquemort is not a phagocytic engulfment receptor for apoptotic cells or bacteria", but the text passage contains no mention of phagocytosis of bacteria, and phagocytosis of bacteria is not tested in the S17 supplement. I would suggest deleting this passage entirely if there is not going to be any discussion of the immune-related phenotypes.

The claim "Toll is not activated by overexpression of GGBP3 or Grass: Experiments performed for ReproSci find that contrary to previous reports, overexpression of GGBP3 (Gottar et al., 2006) or

Grass (El Chamy et al., 2008) in the absence of immune challenge does not effectively activate Toll signaling (Supplementaries S6, S7)" is overly strongly stated unless the authors can directly repeat the original published studies with identical experimental conditions. In the absence of that, the claim in the present manuscript needs to be softened to "we find no evidence that..." or something similar. The definitive claim "does not" presumes that the current experiments are more accurate or correct than the published ones, but no explanation is provided as to why that should be the case. In the absence of a clear and compelling argument as to why the current experiment is more accurate, it appears that

there is one study (the original) that obtained a certain result and a second study (the present one) that did not. This can be reported as an inconsistency, but the second experiment does not prove that the first was an error. The same comment applies to the refutation of the roles for Edin and IRC. Even though the current experiments are done in the context of a broader validation study, this does not automatically make them more correct. The present work should adhere to the same standards of reporting that we expect in any other piece of science.

The statement "Furthermore, evidence from multiple papers suggests that this result, and other instances where mutations have been found to specifically eliminate Defensin expression, is likely due to segregating polymorphisms within Defensin that disrupt primer binding in some genetic backgrounds and lead to a false negative result (Supplementary S20)" should include citations to the multiple papers being referenced. This passage would benefit from a brief summary of the logic presented in S20 regarding the various means of quantifying Defensin expression.

In S22 Results, the statement "For general characterization of the IrcMB11278 mutant, including developmental and motor defects and survival to septic injury, see additional information on the ReproSci website" is not acceptable. All necessary information associated with the paper needs to be included in the Supplement. There cannot be supporting data relegated to an independent website with no guaranteed stability or version control. The same comment applies to "Our results show that eiger flies do not have reduced feeding compared to appropriate controls (See ReproSci website)" in S25.

Supplement S21 appears to show a difference between the wildtype and hemese mutants in parasitoid encapsulation, which would support the original finding. However, the validation experiment is performed at a small sample size and is not replicated, so there can be no statistical analysis. There is no reported quantification of lamellocytes or total hemocytes. The validation experiment does not support the conclusion that the original study should be refuted. The S21 evaluation of hemese must either be performed rigorously or removed from the Supplement and the main text.

In S22, the second sentence of the passage "Due to the fact that IrcMB11278 flies always survived at least 24h prior to death after becoming stuck to the substrate by their wings, we do not attribute the increased mortality in Ecc15-fed IrcMB11278 flies primarily to pathogen ingestion, but rather to locomotor defects. The difference in survival between sucrose-fed and Ecc15-fed IrcMB11278 flies may be explained by the increased viscosity of the Ecc15-containing substrate compared to the sucrose-containing substrate" is quite strange. The first sentence is plausible and a reasonable interpretation of the observations. But to then conclude that the difference between the bacterial treatment versus the control is more plausibly due to substrate viscosity than direct action of the bacteria on the fly is surprising. If the authors wish to put forward that interpretation, they need to test substrate viscosity and demonstrate that fly mortality correlates with viscosity. Otherwise, they must conclude that the validation experiment is consistent with the original study.

In S27, the visualization of eiger expression using a GFP reporter is very non-standard as a quantitative assay. The correct assay is qPCR, as is performed in other validation experiments, and which can easily be done on dissected fat body for a tissue-specific analysis. S27 Figure 1 should be replaced with a proper experiment and quantitative analysis. In S27 Figure 2, the authors should add a panel showing that eiger is successfully knocked down with each driver>construct combination. This is important because the data being reported show no effect of knockdown; it is therefore imperative to show that the knockdown is actually occurring. The same comment applies everywhere there is an RNAi to demonstrate a lack of effect.

The Drosomycin expression data in S3 Figure 2A look extremely noisy and are presented without error bars or statistical analysis. The S4 claim that sphinx and spheroid are not regulators of the Toll pathway because quantitative expression levels of these genes do not correlate with Toll target expression levels is an extremely weak inference. The RNAi did not work in S4, so no conclusion should be inferred from those experiments. Although the original claims in dispute may be errors in both cases, the validation data used to refute the original claims must be rigorous and of an acceptable scientific standard.

In S6 Figure 1, it is inappropriate to plot $n=2$ data points as a histogram with mean and standard errors. If there are fewer than four independent points, all points should be plotted as a dot plot. This comment applies to many qPCR figures throughout the supplement. In S7 Figure 1, "one representative experiment" out of two performed is shown. This strongly suggests that the two replicates are noisy, and a cynical reader might suspect that the authors are trying to hide the variance. This also applies to S5 Fig 3. Particularly in the context of a validation study, it is imperative to present all data clearly and objectively, especially when these are the specific data that are being used to refute the claim.

Other comments:

In S26, the authors suggest that much of the observed melanization arises from excessive tissue damage associated with abdominal injection contrasted to the lesser damage associated with thoracic injection. I believe there may be a methodological difference here. The Methods of S27 are not entirely clear, but it appears that the validation experiment was done with a pinprick, whereas the original Mabary and Schneider study was done with injection via a pulled capillary. My lab group (and I personally) have extensive experience with both techniques. In our hands, pinpricks to the abdomen do indeed cause substantial injury, and the physically less pliable thorax is more robust to pinpricks. However, capillary injections to the abdomen do virtually no tissue damage - very probably less than thoracic injections - and result in substantially higher survivals of infection even than thoracic injections. Thus, the present manuscript may infer substantial tissue damage in the original study because they are employing a different technique.

<https://doi.org/10.7554/eLife.108404.1.sa0>